THE JOURNAL OF PHYSICAL CHEMISTRY A
A JOURNAL OF THE AMERICAN CHEMICAL SOCIETY

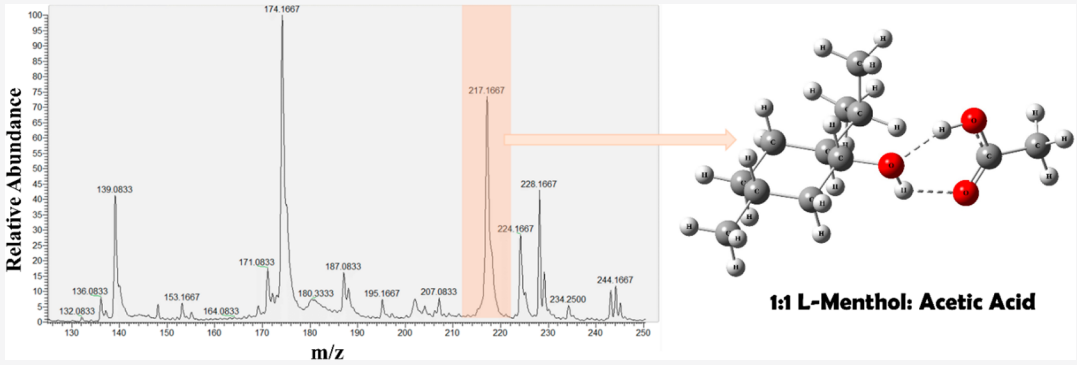
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Structure Elucidation of Menthol-Based Deep Eutectic Solvent using Experimental and Computational Techniques

Md Ackas Ali,[⊥] Md Sajjadur Rahman,[⊥] Ranen Roy, Paul Gambill, Douglas E. Raynie, and Mohammad A. Halim*

Cite This: J. Phys. Chem. A 2021, 125, 2402–2412 Read Online

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ABSTRACT: The structural properties and nonbonding interactions of a menthol-based deep eutectic solvent (DES) were investigated in detail employing experimental and computational methods. A mass spectrometry analysis confirmed the formation of 1:1 L-menthol/acetic acid. A molecular dynamics simulation was used to figure out energetically most favorable cluster conformers of the 1:1 L-menthol/acetic acid system. Density functional theory at the ωB97XD/6-311G (d,p) level of theory was employed to optimize the isolated structures and to calculate their thermochemical properties. Both experimental and computed IR spectra were analyzed for the samples. Additionally, vibrational circular dichroism (VCD) spectra of the samples were measured to prove the chirality transfer. Principal component analysis (PCA) was used to make the data interpretation more vivid. All the spectral data analyses and nanostructure elucidation proved the spontaneous formation of the DES through the formation of strong hydrogen bonding. Experimental solvatochromism and computed highest occupied molecular orbital–lowest unoccupied molecular orbital gaps validated the reasoning. Moreover, comparative VCD and IR spectral analyses clearly indicated a chirality transfer from the chiral menthol to achiral acetic acid. This study suggests that various techniques, such as mass spectrometry, IR, solvatochromism, and computed IR-VCD could be useful and important tools to elucidate nanostructure and nonbonding interactions of a DES. VCD could be used as an excellent complementary technique to IR spectroscopy for a chiral molecule-based DESs.

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INTRODUCTION

A deep eutectic solvent (DES) is a green solvent in which two or more components are mixed at a certain molar ratio and form a eutectic solvent through the depression of freezing points of the components.[1] Commonly a hydrogen-bond acceptor (HBA) and a hydrogen-bond donor (HBD) form a complex nonbonding interaction network, inhibit the process of crystallization, and thus lower the freezing point of the solvent phase beyond those of its individual components.[2,3] Charge delocalization via hydrogen bonds is one of the major reasons behind the freezing point depression.[2] Choline chloride and menthol are the common compounds for the formulations of DESs.[2] Various studies suggested that heat mixing at ∼40 °C or mixing at the ambient temperature with constant stirring are the preferred methods for the formulations of menthol-based DESs.[2,4−6] To prove a DES formation, exploring the impact of temperature, to determine their structures and the chemical properties, mass spectrometry (MS) can be employed.[7,8] Electrospray ionization (ESI) coupled mass spectrometry is well-suited to identify a diverse class of substances including volatile, nonvolatile, polar, and nonpolar small molecules to large biomolecules.[8]

With molecular dynamics (MD) simulation and density functional theory (DFT) calculation, structural orientations of the components in a DES, the calculation of their thermodynamic properties, charge delocalization among HBA

Received: November 30, 2020
Revised: March 9, 2021
Published: March 23, 2021



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https://doi.org/10.1021/acs.jpca.0c10735
J. Phys. Chem. A 2021, 125, 2402–2412

and HBD, and intermolecular nonbonding interactions of DESs can be investigated.[9−12] Ashworth et al. conducted rigorous quantum calculations (QC) on 1:2 choline chloride/urea DES, where they suggested that, along with the other typical interactions, an unconventional doubly ionic hydrogen bond helps the DES to be formed.[13] In addition, QC and MD was employed to analyze the effect of water on the DES structure and the interactions among the DES molecules and protein.[14] Fourier transform infrared (FTIR) spectroscopy is one of the most popular tools to examine intermolecular interactions and the formation of hydrogen bonds in a DES.[15,16] Vibrational circular dichroism (VCD) could be an excellent alternative and complementary technique to IR spectroscopy, when a DES is formed by at least one chiral compound,[17] because VCD signatures are calculated by the difference between left and right circularly polarized IR radiation.[18] Gobi and his co-workers used VCD spectra to prove that a hydrogen bond plays a major role to form a 2-chloropropinoic acid-$CHCl_3$ complex.[19] Kessler et al. stated that DFT is a reliable method to compute VCD spectra and that it can help to illustrate long-range interactions of a protein structure.[20] Principal component analysis (PCA), a popular multivariate dimension-reduction tool, could be used to visualize subtle differences of spectral data of a large number of similar chemical structures.[21−23]

The solvatochromism technique could describe a number of microscopic intermolecular properties of a solvent.[24] A deep eutectic solvent system is difficult to explain only by the typical solvent properties (e.g., dielectric constant or dipole moment) because it possesses many interactions, such as doubly ionic effect, hydrogen bonding, halogen bonding, $\pi-\pi$ interactions, and van der Waals forces.[25−27] Most of the available methods can determine only one major type of the interactions or properties. The solvatochromic technique can determine at least four major properties, such as transition energy (E_T^N), hydrogen-bond donor acidity (α), hydrogen-bond acceptor basicity (β), and polarizability (π^*), just by measuring and comparing the change of the solvatochromic responses of the UV−vis absorbance probes (e.g., Nile Red, 4-nitroaniline, and *N*,*N*-diethyl-4-nitroaniline) dissolved in any solvent.[25,28−30]

Our earlier study using a DFT-based VCD technique proved that this tool could be highly efficient to inspect the nonbonding interactions, specifically hydrogen bonding, in a complex system.[31] We observed that VCD signatures of amino acids (methionine, serine, and glutamine) get intensified or broadened due to their noncovalent or hydrogen-bonding interactions with the solvent molecules.[31] It was also noticed that the chirality of the amino acids can be transferred to achiral solvent molecules. Since many of the DESs are formed with chiral molecules (e.g., menthol and thymol), computed VCD signatures could be an excellent tool to investigate the behavior of nonbonding interactions of that chiral molecule-based DES. Therefore, VCD spectra of the cluster structures of a representative DES system 1:1 L-menthol (Men)/acetic acid (AA)[16] are explored in this study to get complementary spectral data to IR and explain interactions among the chiral and achiral components. In brief, the aims of our study are to elucidate the nanostructure, nonbonding interactions, charge delocalization, and chirality transfer of the 1:1 L-Men/AA DES using a broad spectrum of techniques including mass spectrometry, attenuated total reflectance (ATR)-IR, computed IR and VCD spectroscopy, and solvatochromism.

EXPERIMENTAL DETAILS

Materials. Nile Red, 4-nitroaniline, *N*,*N*-diethyl-4-nitroaniline, dichloromethane (DCM), methanol, and L-menthol were purchased from Acros Organics. Acetic acid (glacial) was purchased from Fisher Chemicals. All the chemicals have purity of at least 99.0%.

Preparation of DES. L-Menthol/acetic acid (1:1) was formulated by mixing anhydrous L-menthol and glacial acetic acid at a 1:1 molar ratio under ambient temperature and pressure for 1 h in a covered beaker with constant stirring less than or equal to 600 rpm until a clear liquid was obtained. Mass spectroscopy, ATR-IR, and solvatochromism experiments were conducted using freshly prepared DES, as its components are highly hygroscopic. Prepared solvents can be kept in the −20 °C freezer for further use.

Mass Spectrometry Method. The mass measurement of the freshly prepared L-Men/AA DES was performed without further dilution. A Finnigan LTQ mass spectrometer (Thermo Fisher Scientific) equipped with a heated electrospray ionization (HESI) ion source was used for all MS experiments. As the prepared DES is highly viscous, very low flow rates of 3−5 μL/min were set using a 250 μL Hamilton syringe. For electrospray ionization parameters, the spray voltage of 3−4 kV was used to achieve the high ion current. The capillary temperature was set to 200 °C, and the sheath gas was set to 10. All experiments were performed with 5 microscan, and the maximal injection time was set to 50 ms. Mass spectra were acquired using a mass range of 100−1000 m/z.

Attenuated Total Reflection Infrared Spectroscopy. A Thermo-Fisher Scientific Nicolet 380 FTIR spectrometer with attenuated total reflectance mode was used for the sample analysis to obtain their spectral data. The spectra of the HBA, HBD, and the studied DES were recorded in the 400−4000 cm^{-1} spectral region at ambient temperature. All signals were recorded with a resolution of 4 cm^{-1} and 290 scans, and the spectra were analyzed using EZ OMNIC software. Additionally, the baseline of sample spectra were corrected, denoised, smoothed, and normalized employing Origin Pro 2020.[32]

Preparation of Cluster Conformers. A molecular dynamics simulation was performed on a 1:1 L-Men/AA solvent system in the gas phase to generate its energetically most favorable cluster conformers. Ten pairs of menthol and acetic acid were considered for performing the MD simulation using the YASARA dynamics program,[33] where the applied force field was AMBER 14.[34] Additionally, MD simulations were performed for the 5 and 20 pairs of menthol-AA. Considering a less complex system, having all interactions and stability lead the 10 pairs of menthol and AA system to be considered as the representative ratio for further calculation for the simplification. Figures S1 and S2 explain how a 10:10 reaction is better to generate cluster structures of the DES. For a short-range van der Waals and coulomb interaction, a cutoff radius 8.0 Å was applied. The particle-mesh Ewald (PME) method was employed to compute the long-range interactions.[35] A periodic boundary condition (cell box 58.81 Å × 58.81 Å × 58.81 Å) and the temperature of 298 K were considered using the NVT ensemble for the simulation. All MD simulations were performed for 50 ns, where snapshots were saved at every 100 ps. The simulations were conducted to generate all possible orientations of the HBA and HBD molecules in the cluster structures of the studied DES and were focused only on generating the most stable cluster structures to

https://doi.org/10.1021/acs.jpca.0c10735
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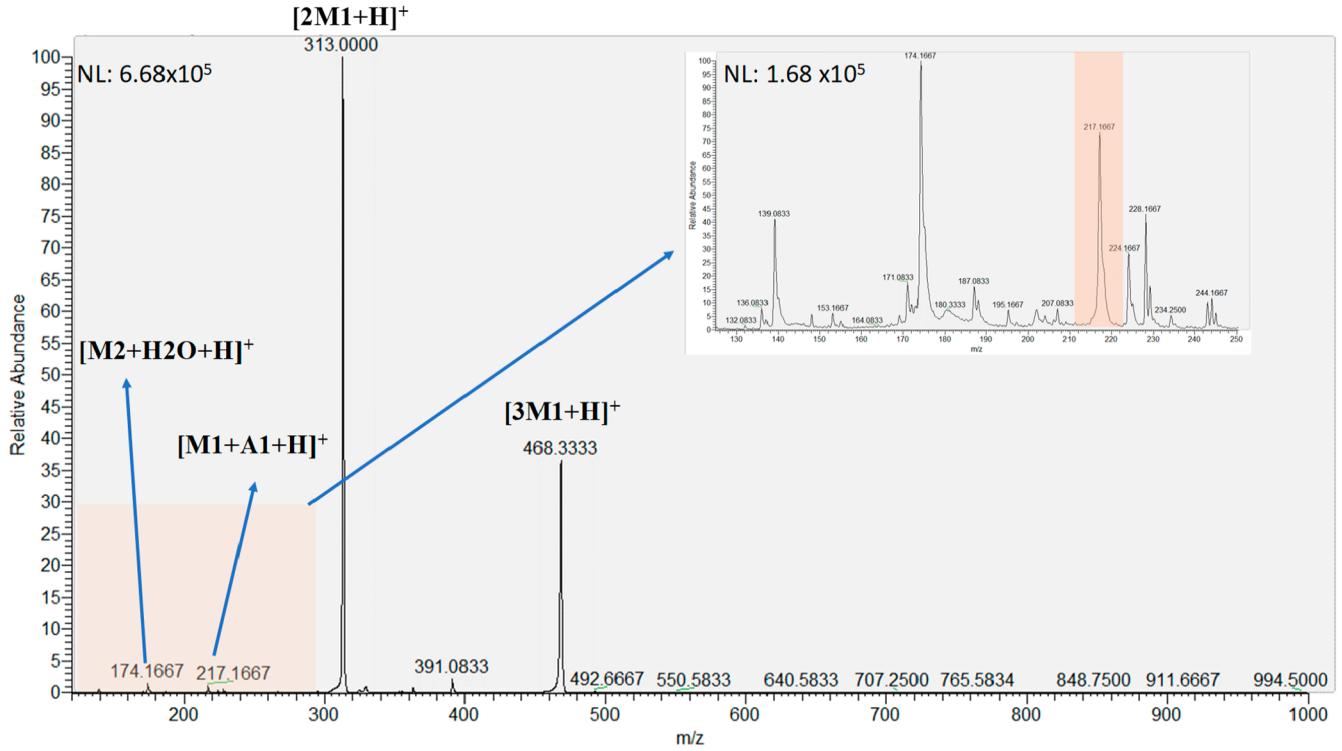


Figure 1. Mass spectrum of menthol/AA DES sample at 1:1 molar ratio at ambient temperature. M = menthol and A = acetic acid.

correlate with the experimental data. It means that this study was done on the simplified model of 1:1 L-Men/AA for interpreting the network of noncovalent interactions that occur in this DES. The most stable cluster conformers of the DES were isolated depending on their single-point energy calculation and distances of nonbonding interactions using the BIOVIA Discovery Studio software package.[36] The isolation of the energetically most favorable clusters was schematically represented in Figure S3.

Computed IR and VCD Spectra Analysis. IR and VCD spectra of all components and the cluster conformers were calculated in the gas phase employing the ωB97XD/6-311G (d,p) level of theory.[27] Although the exact NIST (National Institute of Standards and Technology) vibrational scaling factor was unavailable for this DFT method, all the spectra were multiplied by the scaling factor of 0.957, which is closest to that of the method.[37] Both IR and VCD spectral data were used to examine the most contributing atoms and functional groups to the formation of the studied DES system. This approach added a deeper understanding to the experimental IR spectra and helps to observe the chirality transfer from menthol to acetic acid.

Principal Component Analysis (PCA). PCA reduces the dimensionality of the data to visualize them by identifying their principal source of variation. It reduces the noise and highlights the variabilities present in multivariate spectra of different chemical compounds.[22] The following equation: $X = T_k P_k^T + E$ explains PCA mathematically, where T_k represents how samples relate to each other, P_k signifies how variables relate to each other, k is the number of factors included in the model, and E is the matrix of residuals. All calculations were performed with Origin Pro (2020) equipped with the Principal Component Analysis for Spectroscopy v.1.30 Application.[32] PCA was applied to differentiate the spectral data of menthol, acetic acid, and their cluster conformers. It was also used to compare the efficiency of the VCD over IR technique in terms of distinguishing the subtle differences among the spectral data of the chiral molecule-based DES.

Quantum Mechanical (QM) Calculations and Kamlet−Taft Parameters. The isolated conformers were optimized in the gas phase with the ωB97XD/6-311G (d,p) level of theory. All quantum calculations were performed using the Gaussian 09 software package.[38] Of the four best conformers, only the change of Gibbs free energy of the most spontaneous structure was further calculated in the water phase with the integral equation formalism polarizable continuum model (IEFPCM) using the same level of theory to observe its spontaneity of the structure in a solvent having a high dielectric constant. The electronic energy of this structure is also calculated with augmented double-ζ basis set at the ωB97XD/aug-cc-pVDZ level of theory. Structural and thermochemical properties were calculated from the optimized structures. Highest occupied molecular orbital (HOMO)−lowest unoccupied molecular orbital (LUMO) gaps were also investigated to understand the electron delocalization among the components of the DES. The HOMO−LUMO gap could provide insights of electron transfer or hydrogen-bond transfer among the components. Besides, calculating ΔG, ΔE, ΔH, and ΔS values could add more chemical insights into the hydrogen-bond formation and the spontaneity of the computed clusters. All the computational calculations helped to explain how the cluster structures form or how spontaneous they are, but they could not provide any outcome regarding the extent to which the HBA and HBD molecules accept and donate hydrogen bonds. To get a clear understanding of it, experimental solvatochromism was inevitable for this study.

Solvatochromic measurements were accomplished using three widely used dyes including Nile red, 4-nitroaniline, and

https://doi.org/10.1021/acs.jpca.0c10735
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N,*N*-diethyl-4-nitroaniline. Individual dye solutions of 5×10^{-4} M were prepared in DCM. Aliquots of the dye solutions were micropipetted and loaded to glass vials to mix them well with 1:1 L-Men/AA. A control solution (methanol−dye) was similarly prepared. A 250 μL aliquot of each sample was transferred into a 96-well plate in triplicate. A Synergy H (Biotek) microplate reader was used to determine the wavelengths of maximum absorbances (λ_{max}) of the dyes at 25 °C. Absorbance readings were collected from the samples by a spectral scanning at 1 nm intervals. All solvatochromic properties of the DESs were calculated using the following equations.

$$E_T(\mathrm{NR})_{\mathrm{kcal\cdot mol^{-1}}} = \frac{28\,591.44}{\lambda_{max}(\mathrm{nm})}$$

$$\pi^* = 0.314(27.52 - \upsilon_{N,N\text{-diethyl-4-nitroaniline}})$$

$$\alpha = 0.0649E_T(\mathrm{NR}) - 2.03 - 0.72\pi^*$$

$$\beta = \frac{1.035\upsilon_{N,N\text{-diethyl-4-nitroaniline}} + 2.64 - \upsilon_{4\text{-nitroaniline}}}{2.80}$$

$$E_T{}^N = \frac{E_T(\mathrm{solvent}) - E_T(\mathrm{TMS})}{E_T(\mathrm{water}) - E_T(\mathrm{TMS})}$$

where υ = wavenumber, α = hydrogen-bond donor acidity, β = hydrogen-bond acceptor basicity, π^* = polarizability, and $E_T{}^N$ (unitless) and ET(NR) refer to transition energy. In this study, α and β values of the glacial acetic acid (HBD) and the DES have mainly been focused on.

RESULTS AND DISCUSSION

Results from Mass Spectrometry. The mass spectrometry experiments are performed in the positive mode. A freshly prepared L-Men/AA DES sample at 1:1 molar ratio is directly infused using a syringe pump to the HESI ion source. The mass spectrum of the L-Men/AA DES sample is presented in Figure 1. The exact masses and assignments of the detected peaks are summarized in Table 1. The peak noticed at *m*/*z* 217.1667 can be assigned to L-Men/AA at 1:1 molar ratio. The intensity of this peak is moderately high at 1.46×10^4. The two intense peaks detected at *m*/*z* 313.000 and *m*/*z* 468.3333 correspond to two and three menthol molecules. Besides these peaks, two other peaks noticed at *m*/*z* 139.0833 and *m*/*z* 174.1667 can be assigned to two acetic acid−water and menthol−H_2O complexes. There is no peak related to menthyl acetate observed at ambient temperatures. Despite the highly concentrated sample, no large cluster complex combining L-Men/AA is observed.

Table 1. Experimental and Theoretical *m*/*z* and Assignments of Peaks Detected from 1:1 L-Men/AA DES Prepared at Ambient Temperature

experimental *m*/*z*	theoretical *m*/*z*	assignment[a]
139.0833	139.1267	$[2A1+H_2O+H]^+$
174.1667	174.2796	$[M2+H_2O+H]^+$
217.1667	217.3241	$[M1+A1+H]^+$
313.0000	313.5367	$[2M1+H]^+$
468.3333	468.7932	$[3M1+H]^+$

[a]M1 = menthol (formula mass 156.2644), M2 = deprotonated menthol (formula mass 155.2565), and A1 = acetic acid (formula mass 60.0518).

The effect of temperature on the L-Men/AA DES synthesis is also investigated. When the temperature is raised to 50 °C, no menthol/AA complex at a molar ratio of 1:1 is noticed. Instead, an intense peak is observed at *m*/*z* 221.8333, which can be assigned to menthyl acetate and sodium ion (Figure S4). In addition, a peak is found at *m*/*z* 103.6767 related to acetic anhydride. However, menthyl acetate and acetic anhydride peaks are not present when the DES synthesis is performed at ambient temperature. Interestingly, an intense peak is found at *m*/*z* 356.9167, which is assigned to two menthols with one carbon dioxide. Besides, several peaks associated with one and two menthols are noticed at *m*/*z* 176.2500 and 309.0833, respectively (Figure S1).

ATR-FTIR Spectroscopy. In this study, acetic acid and L-Men act as HBD and HBA, respectively. It was reported that L-Men is an excellent HBA to form the DES with various HBDs even with big molecules like cholesterol.[39] Therefore, experimental ATR-FTIR analyses were performed on L-menthol, acetic acid, and 1:1 L-Men/AA to observe the contributions of the molecules or their functional groups to form the DES. Figure 2 shows the FTIR spectra for L-Men, acetic acid, and 1:1 L-Men/AA, and Table 2 shows the relative shifts of the component's peaks in the spectrum of the DES. It indicates which component contributes more to the formation the hydrogen bonds.

The FTIR spectrum of menthol shows a broad peak at ∼3300 cm^{-1}, which corresponds to its hydroxyl group. Besides, the peaks in the range of 2870−2940 cm^{-1} correspond to an sp^3 C−H bond. In the acetic acid spectrum, a sharp peak is observed at 1700 cm^{-1} for the −C═O of its carboxylic group, and a band for O−H stretch of the same group is found at 3060 cm^{-1}.[40] The spectrum of 1:1 L-Men/AA shows that the peak at 3300 cm^{-1} from the menthol gets broadened due to the H-bond formation with the AA molecules. This spectrum also shows that the carbonyl peak for the eutectic gets sharper and shifts (shown in Table 2) to a higher spectral region (∼1710 cm^{-1}) compared to the same peak of the pure acetic acid (∼1700 cm^{-1}). The higher percentage of peak shifting indicates AA contributes more to the formation of the classical hydrogen bonds.

In the fingerprint region (1500−600 cm^{-1}) of the AA spectrum, a representative C−O stretching band appears at 1300 cm^{-1}. A similar band is found in the spectrum of the DES with a lower intensity and is shifted to a lower spectral region. In addition, peaks for C−H stretching at 1410 and 1460 cm^{-1} were observed for AA and L-Men, respectively. No significant new peak appeared in that hydrocarbon region of the 1:1 L-Men/AA spectrum. Overall, the spectral pattern of the DES correlates more with the AA spectral pattern, and the higher percentage of peak shift of $-OH_{AA}$ suggests that AA strongly contributes for the formation of this DES.

The absorption bands for both L-Men and AA in the region from 1500 to 600 cm^{-1} are difficult to interpret. Therefore, computed IR spectra of the L-Men, AA, and 1:1 L-Men/AA are also analyzed. Since menthol is a chiral molecule, computed VCD spectra of them are studied to get an understanding of the chirality transfer. The stronger contribution of acetic acid to form hydrogen bonds could be further analyzed in detail by the computed IR. Besides, a chirality transfer from the menthol to acetic acid could be proved using their computed VCD spectral data and PCA.

Computed IR and VCD Spectra Analysis. The optimized cluster conformers of the DES show two major

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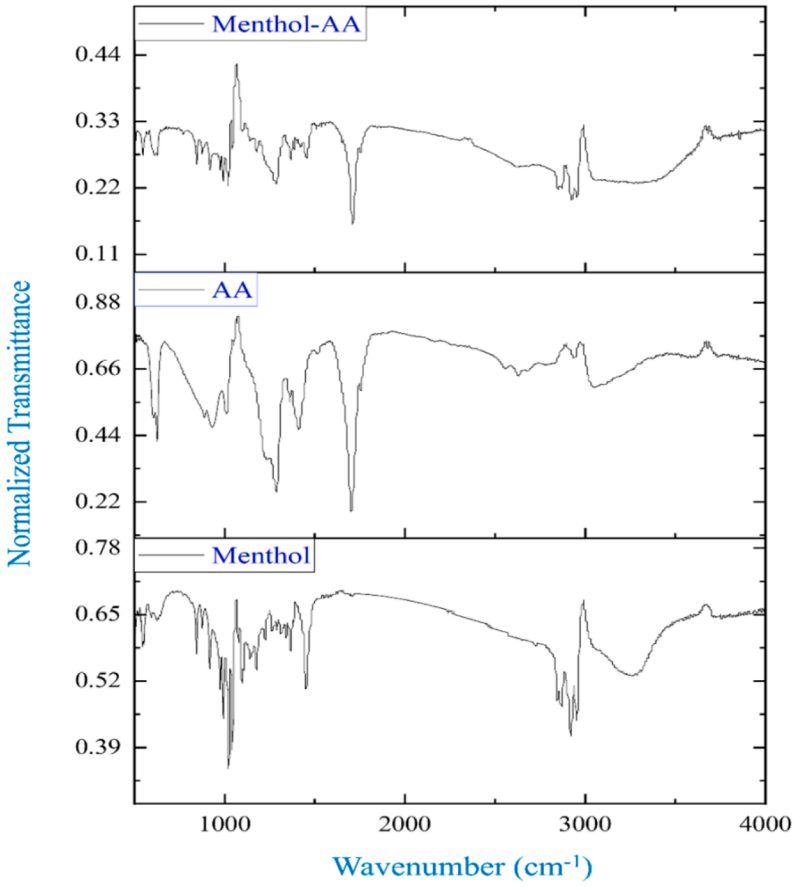


Figure 2. FTIR spectra of L-Men (bottom), acetic acid (middle), and 1:1 L-Men/acetic acid (top).

Table 2. Relative Wavenumber Changes (%) in the Spectrum of the DES

functional group		wavenumber (cm^{-1})		
acetic acid (AA)	1:1 L-Men/AA	acetic acid (AA)	1:1 L-Men/AA	relative changes (%)
C−O stretching	C−O stretching	1300	1290	−0.77
carbonyl (C−O)	carbonyl (C−O)	1700	1710	0.59
−OH	−OH	3060	3300	7.84
DL-menthol	1:1L-Men/AA	L-Men	1:1 L-Men/AA	
carbonyl (C−O)	carbonyl (C−O)		1710	
sp^3 C−H	sp^3 C−H	2870−2940	2870−2960	0.68
−OH	−OH	3290	3300	0.30

hydrogen bonds (Figure 3), $O(20)_{Men}\cdots H(39)-O(38)_{AA}$ and $O(20)-H(21)_{Men}\cdots O(37)_{AA}$, which are mainly responsible for the formation of the DES. In all the cluster conformers, the $O(20)_{Men}\cdots H(39)-O(38)_{AA}$ hydrogen-bond distances range from 1.72 to 1.77 Å, and the $O(20)-H(21)_{Men}\cdots O(37)_{AA}$ distances range from 2.01 to 2.13 Å. Therefore, it can be inferred that $O(20)_{Men}\cdots H(39)-O(38)_{AA}$ is the strongest hydrogen bond in all the structures. Because of the formation of the DES, some of the covalent bond distances and bond angles of menthol and acetic acid have changed in the cluster conformers (Figure 3 and Table S2 in the Supporting Information). Values of ΔG (change in Gibbs free energy) and ΔH (change in enthalpy) for all conformers (Table 3) confirm that the formations of the DES clusters are spontaneous and that the hydrogen bonds are strong enough to elongate the covalent bonds, such as O(7)−H(8) of acetic acid and O(1)−H(31) of menthol (Table 3 and Figure 3).

IR and VCD spectra can prove the hydrogen bonds and the elongations of the covalent bonds. Calculated IR and VCD spectra and the spectral data of L-Men, acetic acid, and four cluster conformers of the DES in the gas phase are presented in Figure 4 and Table S1 of the Supporting Information. Menthol has an −OH functional group bonded with a chiral carbon, and it shows a C−O stretching vibration in the range of 1024−1056 cm^{-1}.[41] In the computed menthol spectrum, the C(3)−O(1) stretching vibration has been shown at 1019 cm^{-1}. So, computed data show a close agreement with the experimental data. Between 1258 and 1455 cm^{-1} of the spectrum, C−H bending vibrations appear.[41] In the spectra of the cluster conformers, weak peaks for C−H bending vibrations appear in the same region. That means that all the C−H peaks of menthol and AA in that region have been merged together and gotten smaller. It is an indication of the formation of the studied DES.

Figure 4a and Table S1 show that all the cluster conformers have C−H stretching vibrations at 1368−1406 cm^{-1} for menthol. Besides, conformers B and C show peaks at 657 and 684 cm^{-1}, respectively, for the $-O(38)-H(39)_{AA}$ bending vibration. Morino et al. illustrated that methyl and methylene groups of menthol also have strong C−H vibrations in the 2848−2957 cm^{-1} range. Similar C−H peaks are observed in the calculated menthol spectrum in the range of 2800−3000 cm^{-1}. Moreno et al. reported that menthol shows a weak −OH

https://doi.org/10.1021/acs.jpca.0c10735
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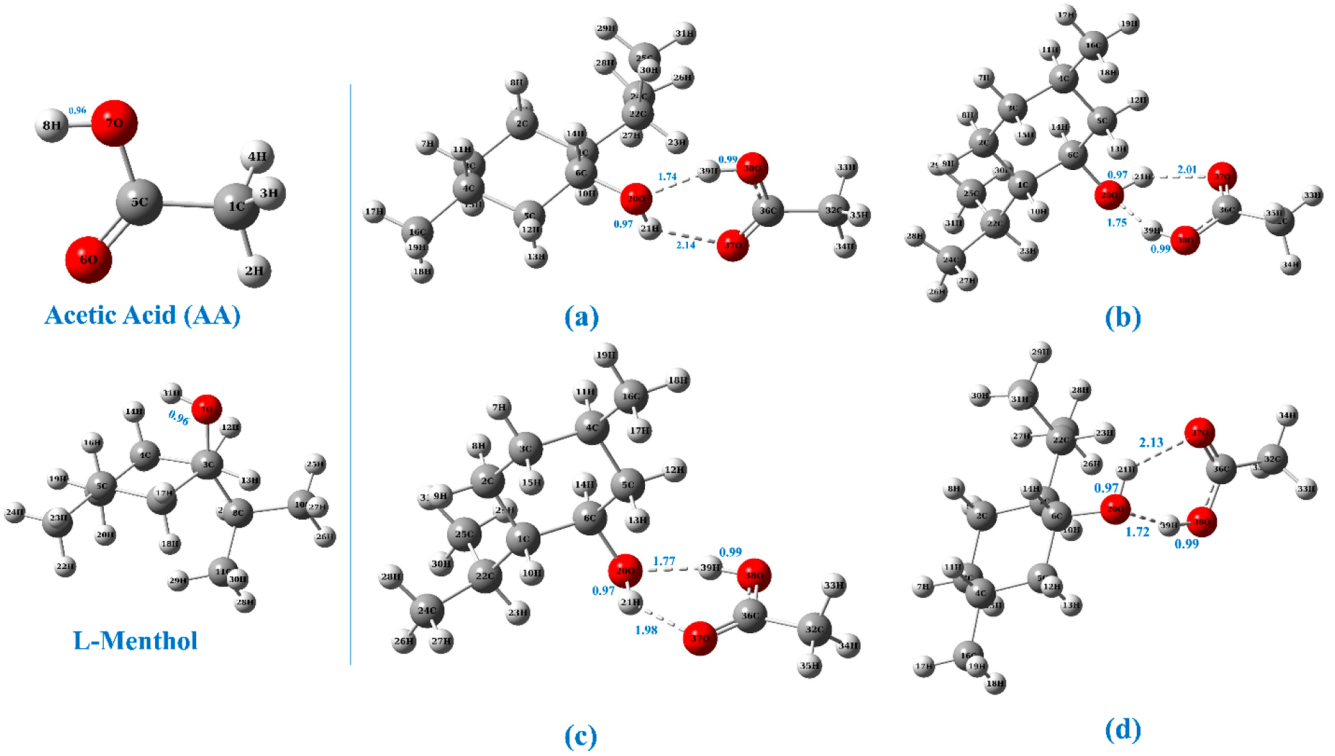


Figure 3. Energetically most favorable cluster conformers of 1:1 L-Men/AA DES are (a) conformer A, (b) conformer B, (c) conformer C, and (d) conformer D, which are optimized at the ωB97XD/6-311G (d,p) level of theory, and the bond distances are in angstroms. AA and L-Men optimized structures are also given in this figure.

Table 3. HOMO−LUMO Energy Gap (values in eV), Δ*G*, Δ*E*, Δ*H* (in kJ/mol), Δ*S* (in J/mol/K) of Menthol, Acetic Acid, and the DES's Conformers Calculated using the ωB97xD/6-311G(d,p) Level of Theory in the Gas Phase and with Standard Conditions

parameter	menthol	AA	conformer A	conformer B	conformer C	conformer D
LUMO	−1.2	−2.83	−2.83	−2.82	−2.82	−2.84
HOMO	−9.82	−9.95	−9.82	−9.72	−9.81	−9.73
HOMO−LUMO gap (eV)	8.62	7.12	6.99	6.9	6.99	6.89
Δ*G* (kJ/mol)			−18.43	−16.87	−15.20	−22.24
Δ*E* (kJ/mol)			−56.12	−55.67	−54.27	−58.97
Δ*H* (kJ/mol)			−58.60	−58.14	−56.75	−61.44
Δ*S* (J/mol/K)			−134.77	−138.50	−139.43	−131.57

stretching vibration at 3644 cm^{-1}.[42] In this study, a weak −O(1)−H(31)$_{Men}$ stretching vibration appears at 3741 cm^{-1}.

In the AA spectrum, the C(5)−O(7) stretching vibration is observed at 1183 cm^{-1}, and C(5)═O(6) is seen at 1800 cm^{-1}. C(36)═O(37) stretching vibrations in the spectra of the cluster conformers are found between 1745 and 1748 cm^{-1}. A weak peak for the −O(38)−H(39) functional group is detected at 3679 cm^{-1}, but the peak gets amplified and shifted to the lower wavenumber (3231−3263 cm^{-1}) in the spectra of the cluster conformers (Figure 4a). It has also been observed that the peaks that appear for O(38)−H(39) of AA in the conformer's spectra are more intense than the peaks corresponding to the O(20)−H(21) of the pure menthol. It happens because the −OH part of AA makes a stronger hydrogen bond than the −OH functional group of the menthol. To gain a better insight, it would be better to compare the bond distances of the O−H in menthol, acetic acid, and their DES clusters. When menthol and acetic acid are in their pure forms, bond distances of O(10)−H(31)$_{Men}$ and O(7)−H(8)$_{AA}$ are the same (0.96 Å). But when menthol and acetic acid form H-bonds between them, the O(38)−H(39)$_{AA}$ bond gets more elongated (0.99 Å) than the O(20)−H(21)$_{Men}$ (0.97 Å) bond in all four cluster conformers (Figure 3). It proves that the stronger H-bonding appears as a more intense peak in the spectra. Therefore, the peak for O(20)$_{Men}$···H(39)−O(38)$_{AA}$ appears stronger than the peak for O(20)−H(21)$_{Men}$···O(37)$_{AA}$.

Menthol is an −OH group-containing monoterpene. In Figure 4b, the VCD spectrum of L-Men has a peak for the O(1)−H(31) bending vibration, and the rotational strength is 16.12 × 10^{-44} esu^2 cm^2. A negative rotational strength of −8.02 × 10^{-44} esu^2 cm^2 is obtained for the C(3)−O(1) stretching vibration. In Table S1, rotational strength values of O(20)−H(21)$_{Men}$ bending vibrations are (90.68, −107.16, and −23.87) × 10^{-44} esu^2 cm^2 for conformers B, C, and D, respectively, where the value for conformer A is 3.91 × 10^{-44} esu^2 cm^2. Also, the O(1)−H(31) stretching vibration of menthol shows weak VCD rotational signal at 3741 cm^{-1},

https://doi.org/10.1021/acs.jpca.0c10735
J. Phys. Chem. A 2021, 125, 2402−2412

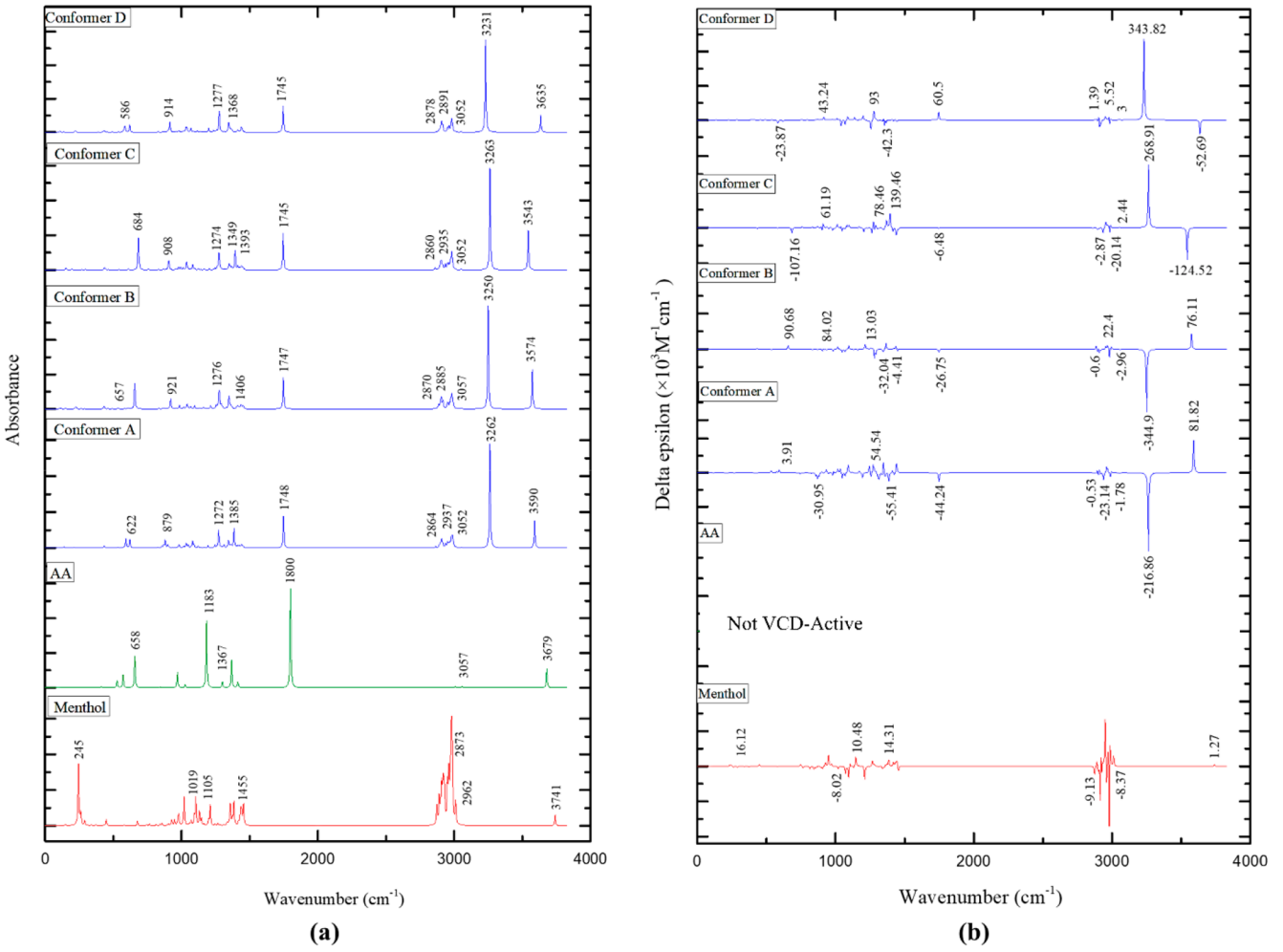


Figure 4. Calculated (a) IR spectra and (b) VCD spectra of L-Men, acetic acid (AA), and the cluster conformers (labeled as A, B, C, and D) of 1:1 L-Men/AA, where AA is not a VCD active compound.

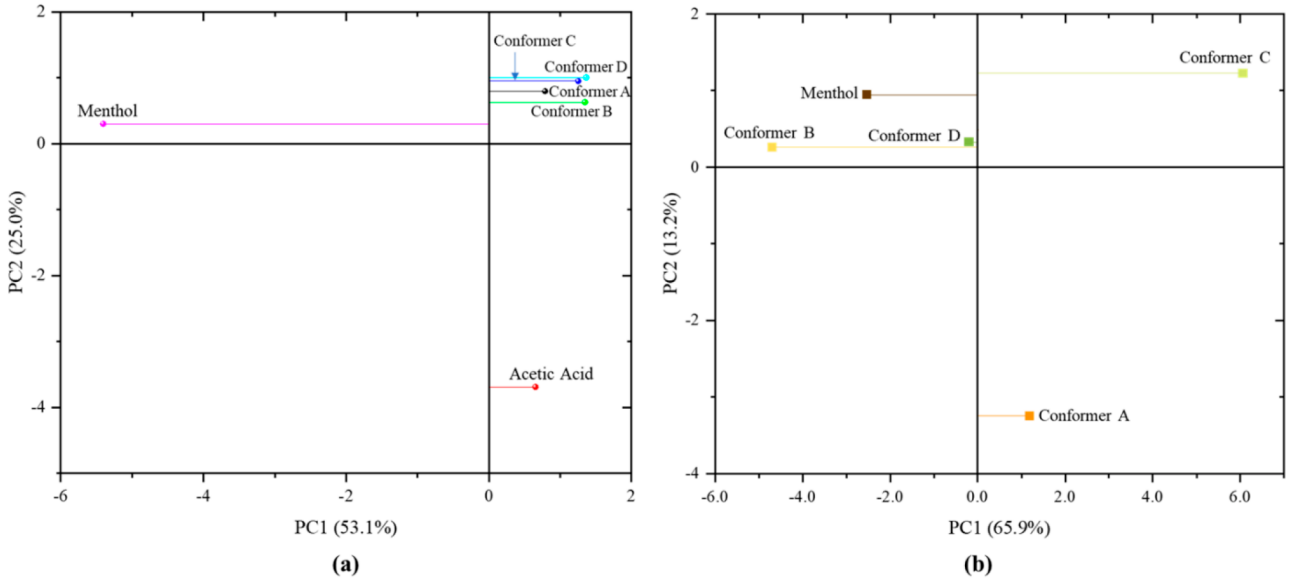


Figure 5. Score plots from PCA of the (a) IR and (b) VCD simulated spectral data (400−2000 cm^{-1}), specifically to compare complex fingerprint regions of the DES cluster conformers.

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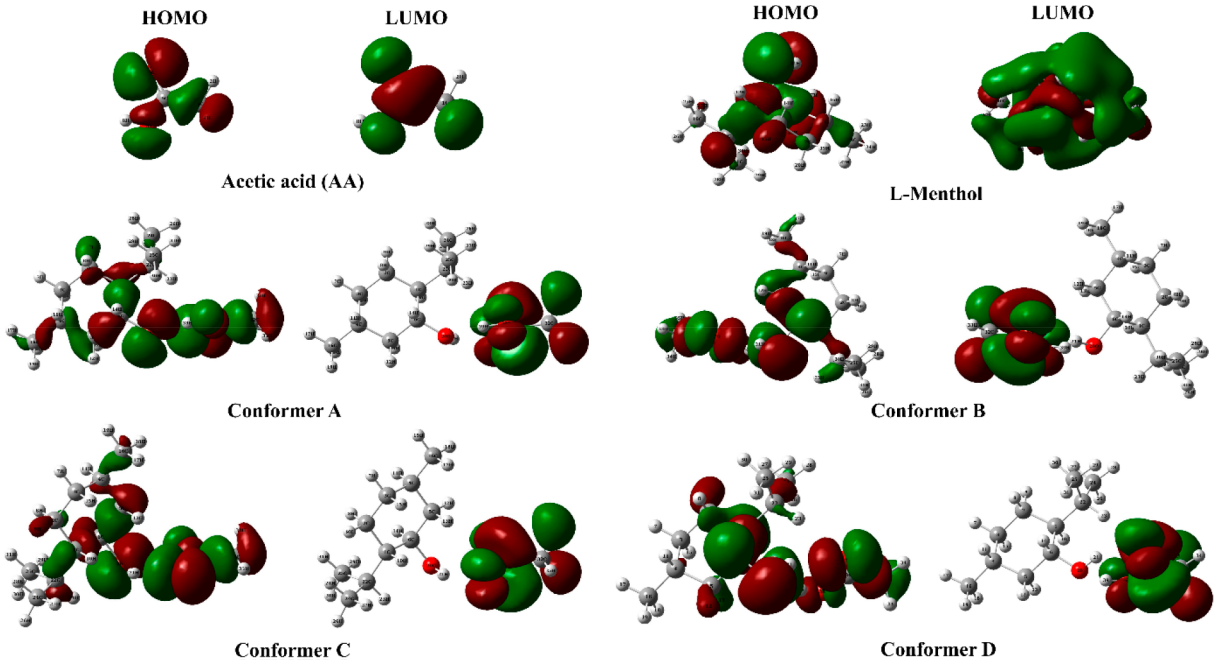


Figure 6. Selected frontier molecular orbitals (HOMOs and LUMOs) of menthol, acetic acid, and conformers calculated at the ωB97XD/6-311G (d,p) level of theory in the gas phase.

which is 1.27 × 10^{-44} esu^2 cm^2. In the VCD spectra of the conformers (Figure 4b), the −OH peaks get intensified due to the strong hydrogen bonding,[31] and their rotational strengths are 81.82 × 10^{-44} esu^2 cm^2 (conformer A), 76.11 × 10^{-44} esu^2 cm^2 (conformer B), −124.52 × 10^{-44} esu^2 cm^2 (conformer C), and −52.69 × 10^{-44} esu^2 cm^2 (conformer D).

Acetic acid is not a VCD active molecule, but in the spectra of the conformers a few strong peaks have been observed for AA. Values of the rotational strengths in spectra of conformers A, B, C, and D are 54.54 × 10^{-44}, 13.03 × 10^{-44}, 78.46 × 10^{-44}, and 93.00 × 10^{-44} esu^2 cm^2, respectively, for the C(36)−O(38) stretching vibration, and 30.95 × 10^{-44}, 84.02 × 10^{-44}, 61.19 × 10^{-44}, and 43.24 × 10^{-44} esu^2 cm^2, respectively, for the O(38)−H(39) bending vibration. These stronger positive and negative bands in the conformer's spectra than in the menthol spectrum represent the formation of hydrogen bonding and chirality transfer. Because of the formation of $O(20)_{Men}\cdots H(39)-O(38)_{AA}$ and $O(20)-H(21)_{Men}\cdots O(37)_{AA}$ hydrogen bonds, O(38)−H(39) stretching vibrations of AA for the conformers show highly strong responses, −216.86 × 10^{-44}, −344.9 × 10^{-44}, 268.91 × 10^{-44}, and 343.82 × 10^{-44} esu^2 cm^2, respectively.

In the L-Men spectrum, one positive band having a weak rotational signal was observed over 3200 cm^{-1}. But in that region two strong peaks are emerged in all the spectra of the conformers. A stronger peak appears in the lower wavenumber range (3200−3300 cm^{-1}), and the other one appears in the higher wavenumber range (3500−3800 cm^{-1}). In all cases, one of them is a positive band, and another one appears as a negative band. For example, in the spectrum of conformer D, the VCD signal at the lower wavenumber range has a positive rotational strength, which is −216.86 × 10^{-44} esu^2 cm^2. At the higher wavenumber peak, a positive response is shown, and the value is 81.82 × 10^{-44} esu^2 cm^2. These intensified rotational strengths clearly show the formations of $O(20)_{Men}\cdots H(39)-O(38)_{AA}$ and $O(20)-H(21)_{Men}\cdots O(37)_{AA}$ hydrogen bonds.

Principal Component Analysis (PCA). Principal component analysis was performed on the 400−2000 cm^{-1} spectral data of all IR and VCD spectra. The first principal component (PC1) explains the highest possible variability. In Figure 5, the PC1 of IR and VCD spectra explains 53.1% and 65.9% of the spectral data, respectively. Then the PC2 accounts for the second-largest variance of the spectral data, and it explains 25.0% and 13.2%, respectively. Therefore, PC1 and PC2 together illustrate the distribution of IR (Figure 5a) and VCD (Figure 5b) spectra of the pure components (L-Men and acetic acid) and the four DES cluster conformers over 78% on the two PC score axes.

It is observed that the PC1 can successfully separate scores of menthol and acetic acid for IR spectral data (Figure 5a). It is logical, as their spectral features are distinctive from each other. Scores of the four conformers reside close to each other and away from menthol and acetic acid, which confirms that the spectral features of these four conformers are the combination of menthol and acetic acid. Although the PCA of IR spectral data successfully separated menthol and acetic acid from their cluster conformers, it is unable to differentiate the PCA scores of the conformers (Figure 5a). PC1 can explain a higher percentage of spectral data because it is driven by C−O, C═O, and C−H stretching vibrations of the hydrocarbon region, where PC2 is driven by mainly C−O and C═O stretching vibrations. Figure S5 shows a loading plot of the principal component analysis of IR spectral data. Here, C−O and C═O frequencies have large and opposing effects on the PC1 and PC2 scores. Additionally, PC1 is driven by some peaks with a large number of opposing peaks in the hydrocarbon region. These are for C−H stretching peaks of menthol and acetic acid. In the region between 586 and 684 cm^{-1}, the $O-H_{Men}$ bending vibration has also little effect on both the PC scores.

However, the PCA scores generated from VCD spectral data can distinguish the conformers, because the VCD technique is selectively sensitive for the chiral menthol molecule. Although AA is not a VCD active compound, rotational strengths of the

https://doi.org/10.1021/acs.jpca.0c10735
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C−O stretching in the cluster conformers A, B, C, and D are 54.54×10^{-44}, 13.03×10^{-44}, 78.46×10^{-44}, and 93.00×10^{-44} esu^2 cm^2, respectively. Since the peaks correspond to the C−O stretching vibrations of AA, chirality has been transferred from the chiral menthol to achiral AA. Therefore, these clusters are well-structured and show characteristics of a single complex chiral molecule. Therefore, the PCA analysis helped to visualize the similarity and dissimilarity of IR and VCD spectral data of menthol, AA, and their DES conformers. It proves that VCD could be used as a tool that is complementary to IR spectroscopy to investigate a chiral molecule-based DES.

QM Calculation and Kamlet−Taft Parameters Analysis. Comparative HOMO−LUMO gap values of the components and the cluster conformers could help to get hints about how the charge delocalization occurred from one component through another component.[43] A lower HOMO−LUMO gap means a higher chemical reactivity of a compound.[44] HOMO−LUMO gap values of menthol, AA, and conformers A, B, C, and D are 8.62, 7.12, 6.99, 6.90, 6.99, and 6.89 eV, respectively (Table 3 and Figure 6). HOMO−LUMO gaps get decreased in the cluster conformers compared to the HOMO−LUMO gaps of their components, which indicates that the cluster structures are formed by the orbital overlapping or by a charge delocalization from menthol to acetic acid and vice versa. It could also facilitate a chirality transfer from the chiral menthol to the achiral acetic acid through their complex hydrogen-bond network.[45,46] The change in the Gibbs free energy (ΔG), electronic energy (ΔE), and entropy (ΔS) of the clusters were also calculated in the gas phase using the ωB97xD/6-311G(d,p) level of theory (Table 3). ΔG values show that, among the clusters, conformer D is the most spontaneous structure. Besides, ΔS values suggest that the cluster structures are spontaneous even at every low temperature to high temperature.

Since conformer D is the most spontaneous and it shares maximum characteristics of its components, the ΔG value of this structure was also calculated in a high dielectric constant medium (water) using the IEFPCM solvation model at the same level of theory. The ΔG value is found as −7.94 kJ/mol in the water phase. This is just for one orientation of the 1:1 Men/AA structure, and this cannot show the overall scenario. It is well-known that, when water is added to a certain extent to a hydrophobic DES, water molecules start to dominate the hydrogen bond formation.[24] However, a small amount of water could be added to this DES to get more intense and sharper spectroscopic peaks. A higher-level basis set (aug-cc-pVDZ) is also used to calculate the electronic energy of conformer D to obtain a more corrected electronic energy of the cluster. The ΔE value is calculated as −55.25 kJ/mol using the ωB97xD/aug-cc-pVDZ level of theory in the gas phase. Since this basis set is computationally too expensive and the water phase calculation for DES cannot mimic the experimental situation, all the calculations of this study were conducted using the ωB97xD/6-311G(d,p) level of theory in the gas phase.

A hydrophobic deep eutectic solvent is a complex system and does not behave as a typical solvent. It is difficult to picturize the overall scenario of the solvent only by the quantum calculations. To make more sense of this system, an experimental solvatochromism technique has been employed. Kamlet−Taft parameters including α, β, and π^* can excellently explain a DES system.[27] Classical hydrogen bonds play the major role for the depression of the freezing point of the studied solvent. Measured solvatochromic parameters of methanol (control solvent) show a high degree of agreement with the literature values, which proves the reliability of the experiment. The transition energy and polarizability of the methanol are found as 54.98 ± 1.07 and 0.75 ± 0.01 kcal·mol^{-1}, where the literature values are 52.02[47] and 0.77[48] kcal·mol^{-1}, respectively.

Kamlet−Taft parameters, such as α and β of AA and 1:1 L-Men/AA, are mainly investigated from this experiment. Both α and β values were increased for the DES compared to glacial acetic acid (AA). The values are 1.14 and 0.43 for AA and 1.61 and 0.61 for the DES, respectively. The values proved that both the hydrogen-bond donation and reception were increased to a great extent among the DES components compared to the pure AA. These results indicate the formation of this DES is mainly driven by the classical hydrogen bonding, where AA dominates the formation. These values also suggest a higher transition energy for the DES than the AA. The dimensionless E_T^N (transition energy) value of the DES is found as 0.97, whereas for AA it is 0.78. A higher transition energy represents a lower polarizability of the DES. Polarizability values of the DES and AA are 0.55 and 0.65, respectively, which means an electron cloud among the components is well-distributed and localized compared to the glacial AA.

CONCLUSION

In summary, together experimental mass spectrometry, IR, solvatochromism, and computational approaches elucidated the structural features of the studied 1:1 L-Men/AA DES system. A mass spectrometry measurement confirms the formation of the 1:1 L-Men/AA DES system in ambient temperature. However, such a system is not observed when the temperature is elevated to 50 °C during the formulation. The most stable cluster conformers of the DES are figured out using an MD simulation followed by a DFT calculation approach. By monitoring the structures and vibrational spectra, it can be concluded that the major nonbonding interactions among menthol and acetic acid are $O(20)_{Men}\cdots H(39)-O(38)_{AA}$ and $O(20)-H(21)_{Men}\cdots O(37)_{AA}$, which are mainly responsible for the formation of the studied DES. Radial distribution function calculations showed that, along with the classical hydrogen bonds, nonclassical hydrogen bonds (e.g., C−H···O) from alkyl tails of the menthol and acetic acid help to form a strong network throughout the DES system and help to delocalize the charge cloud from one component to another one. VCD spectroscopy followed by PCA is an excellent approach to analyze the nonbonding interactions and chirality transfer of a chiral molecule-based DES. These insights could be useful to study the emerging research areas, such as amino acid-based DES[3] and DES−protein interactions.[49]

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jpca.0c10735.

IR and VCD wavenumbers and intensities, selected bond distances (Å) and bond angles (deg), images of the 5, 10, and 20 pairs of menthol−acetic acid MD simulations, van der Waals and Coulombic energies of the menthol−acetic acid simulations; schematic flowchart shows how the best conformers were selected, mass spectra of this DES when prepared at 50 °C,

https://doi.org/10.1021/acs.jpca.0c10735
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loading plot of the PCA analysis of IR and VCD spectra (PDF)

AUTHOR INFORMATION

Corresponding Author

Mohammad A. Halim − *Department of Physical Sciences, University of Arkansas-Fort Smith, Fort Smith 72913-3649, Arkansas, United States;* orcid.org/0000-0002-1698-7044; Phone: 1-479-788-7741; Email: mohammad.halim@uafs.edu; Fax: +1-479-424-6741

Authors

Md Ackas Ali − *Division of Quantum Chemistry, The Red-Green Research Center, Tejkunipara, Tejgaon, Dhaka 1215, Bangladesh; Department of Chemistry, Jahangirnagar University, Savar, Dhaka 1342, Bangladesh;* orcid.org/0000-0002-8235-3517

Md Sajjadur Rahman − *Department of Chemistry and Biochemistry, South Dakota State University, Brookings 57007, South Dakota, United States;* orcid.org/0000-0002-2996-4845

Ranen Roy − *Department of Chemistry and Biochemistry, South Dakota State University, Brookings 57007, South Dakota, United States*

Paul Gambill − *Department of Physical Sciences, University of Arkansas-Fort Smith, Fort Smith 72913-3649, Arkansas, United States*

Douglas E. Raynie − *Department of Chemistry and Biochemistry, South Dakota State University, Brookings 57007, South Dakota, United States;* orcid.org/0000-0003-1475-8130

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.jpca.0c10735

Author Contributions

⊥M.A.A. and M.S.R. have equal contribution.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

All authors are grateful to the Department of Chemistry and Biochemistry and the Department of Pharmaceutical Science, South Dakota State University, for providing chemicals and the instrumentation support.

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